

EVN HyperLogistics

RVCC Venture Competition Application

Venture: EVN HyperLogistics
Product: HyperLogistics — Safety-compliant, explainable AI rerouting for middle-mile logistics
Date: March 15, 2026
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Demo Video: <https://youtu.be/iZP4qxMR6w4>

Problem & Value Proposition

What problem are you solving and why does your solution improve the situation?

Middle-mile logistics managers face a critical daily problem: when disruptions hit — weather events, traffic accidents, road closures — they must make immediate rerouting decisions that are both operationally sound and physically compliant with DOT regulations. Existing tools can flag risk, but they stop short of producing a safe, explainable reroute recommendation.

We call this the **prediction-action gap**.

The scale of this problem is real. Our system is grounded in 7.7 million US accident records, multi-terabyte NOAA weather data, and 600,000+ DOT bridge constraint records. These are not model assumptions — they define the actual operating environment freight teams work in every day.

HyperLogistics closes this gap by combining three components into a single decision flow:

Component	Role
ReMindRAG	Retrieves grounded evidence for reroute reasoning from historical disruption knowledge
SRSNet	Forecasts short-horizon (4–8 hour) disruption propagation aligned to transit windows
Symbolic Safety Veto	Hard-blocks any route that violates DOT bridge clearance or load limits

The result: a dispatch recommendation that is not just intelligent, but **compliance-ready and explainable**. Every output cites its reasoning and has passed a deterministic safety check — something no general-purpose AI tool currently provides for this workflow.

Customer & Market Opportunity

Who is the target customer and why is this opportunity meaningful?

Primary users: Logistics Network Managers and Area Managers at trucking carriers and third-party logistics providers (3PLs).

These operators own middle-mile execution outcomes. They are accountable for on-time performance, cost efficiency, and safety compliance under constant disruption pressure. Their core job

during a disruption is to triage, reroute, and confirm DOT compliance — often in minutes.

Why this opportunity is meaningful:

- Disruptions are high-frequency and consequential. A single bad reroute can violate bridge load limits, incur DOT liability, or delay a full shipment lane.
- Most routing AI tools today optimize for efficiency but **lack deterministic safety enforcement**. They are not built for compliance-critical decisions.
- HyperLogistics sits at the exact intersection where this is unaddressed: **operations + explainability + hard compliance enforcement**.

To validate product-market fit, we built 100 domain-adapted instruction scenarios drawn directly from real manager decision patterns — disruption queries, bridge constraint vetoes, and reroute approval flows. These are not synthetic edge cases. They reflect the actual language and decisions logistics managers face daily.

Market opportunity:

Layer	Market	2025 Size	Growth
TAM	Middle-Mile Logistics	\$110B	7% CAGR → \$218B by 2035
SAM	Route Optimization Software	\$7.9B	13.3% CAGR → \$16.8B by 2031
SOM	Explainable AI in Logistics	\$1.7B	33.7% CAGR → \$31.7B by 2035

EVN targets the SAM/SOM intersection — the fastest-growing segment (33.7% CAGR) within an active purchasing market (\$7.9B) where compliance-enforcing, explainable AI does not yet exist as a category product.

Sources: ResearchAndMarkets / 360iResearch (TAM), Mordor Intelligence (SAM), Future Market Insights (SOM)

Progress & Learning

What steps have you taken so far and what have you learned?

What We Built

Milestone	Evidence
Live deployment	HyperLogistics Streamlit app is publicly accessible at cs5542hyperlogistics.streamlit.app
Snowflake-native pipeline	Medallion architecture (Bronze/Silver/Gold) with Snowpipe, External Tables, and Spatial SQL
Domain adaptation (Week 8)	Accuracy improved from 40% to 95% using QLoRA fine-tuning (Phi-2, LoRA rank=16) and few-shot prompting
Agent execution (Week 6)	Gemini 2.5 Flash agent with 9 analytics tools — 4/5 scenarios passed (80%), 6/7 tool selections correct (86%)
Reproducibility (Week 7)	ReMindRAG audit: 17/17 automated tests passed , 14/14 identified issues resolved

What We Learned

The biggest barrier to enterprise trust is not model accuracy — it is explainability and compliance enforcement.

Early iterations produced accurate recommendations that managers still could not use, because the system could not cite why a route was chosen or confirm it was physically valid. Once we added the symbolic bridge-constraint veto and the ReMindRAG reasoning trace, confidence in the outputs changed significantly.

The +55 percentage-point accuracy improvement from baseline to adapted model confirmed that domain-specific instruction tuning on logistics jargon and DOT constraint reasoning is not optional — it is the core product differentiator.

Team & Commitment

Who is on the founding team and why are you working on this venture?

Member	Role	Stack Ownership
Tony Nguyen	ML / Full-Stack Engineer	Pipeline automation, model adaptation (QLoRA), S3/Snowflake orchestration, reproducibility
Daniel Evans	Data / Backend Engineer	External data engineering (NOAA weather), Streamlit app delivery, evaluation support
Joel Vinas	Data / ML Engineer	Internal data engineering, ingestion scripts, silver-layer preprocessing

All three are MS Data Science & Analytics students with equal equity and full-stack ownership. We are not advisors with slide decks — we are the engineers who built every component in this system. Responsibilities map directly to code and architecture ownership across the full stack.

Why we are working on this:

We identified the prediction-action gap firsthand while building logistics AI systems for coursework. The constraint-compliance problem is not academic — it is a live operational risk for every freight company running middle-mile operations. We kept building because the technical path was clear and the operating need was real.

We are committed to this venture because the hardest parts are already built: the data pipeline, the retrieval architecture, the safety veto, the adapted model, and the live application. What remains is go-to-market execution.

RVCC Fit

How will RVCC help you move the venture forward?

RVCC fills the precise gaps that technical founders face after building a working product but before achieving first commercial traction.

What We Are Asking For

Mentorship — go-to-market for logistics operations buyers

We understand the technology. We need guidance on how to sell into operations leadership at carriers and 3PLs: who champions this internally, what procurement looks like at mid-size carriers, and how to structure a paid pilot agreement.

Network — freight operator and ecosystem introductions

A warm introduction to one logistics operator, regional carrier, or Snowflake ecosystem partner would compress our sales cycle significantly. The product is live and demonstrable today.

Funding — runway for hardening and first pilot

Specific near-term needs:

- Complete evaluation hardening and lock benchmark reporting
- Scale adaptation dataset beyond 100 examples for broader coverage
- Add enterprise access controls (OAuth / RBAC) required for production deployment
- Support a first paid pilot engagement

Why RVCC Resources Convert Directly to Traction

Current Asset	RVCC Unlock
Live deployed application	Pilot-ready with one enterprise introduction
95% adapted accuracy	Publishable benchmark with mentorship on framing
17/17 reproducibility tests	Engineering credibility for enterprise procurement evaluation
9-tool analytics foundation	Expansion path for enterprise analytics add-on revenue

RVCC at this stage would convert proven technical capability into a go-to-market motion — which is exactly the gap the competition is designed to close.

EVN HyperLogistics — RVCC 2026 Submission